# The Cholestatic Infant: the 2017 North American and European Algorithm, and What Must Be Excluded on Day One.

Picture two infants who arrive in the same clinic on the same afternoon, both three weeks old and both yellow. The first is a breastfed boy who is feeding well, gaining weight and looks entirely comfortable. His stools are light yellow. The second is a girl who has been vomiting after feeds for two days, is sleepier than her mother would like, and has a large, firm liver. Both have a raised conjugated bilirubin. The first infant may well have biliary atresia, and his problem is measured in weeks. The second infant may have bacterial sepsis, galactosemia, or adrenal and pituitary failure, and her problem is measured in hours. The same laboratory number sits in front of us, but the two infants are on very different clocks.

The clinical question of this chapter is how to evaluate a cholestatic infant in the right order, and what must be excluded on the very first day. The main idea is this. The 2017 joint guideline of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, is written mostly to find biliary atresia quickly. But inside it sits a second, quieter rule. A small group of causes, mainly sepsis and urinary infection, galactosemia, hypothyroidism, and hypopituitarism, together with the bleeding risk of vitamin K deficiency, can kill or injure an infant within days, and every one of them is treatable. Those infants usually look ill. Infants with biliary atresia usually look well. So the work-up is ordered by which clock runs fastest, not by which disease is most common.

We will move through the topic in order. First, the definition of cholestasis and when to test. Second, the ten recommendations of the guideline and the order of steps. Third, the first-day tests and what each one is looking for. Fourth, each of the treatable emergencies in turn. Fifth, the imaging and tissue tests that confirm or exclude biliary atresia, and why the clock matters so much. Sixth, how the North American and European documents differ. We finish with the common traps, and with what to do differently in clinic.

A note on sources. The central document is the 2017 guideline on the evaluation of cholestatic jaundice in infants, written by Fawaz and colleagues for the two societies and published in the Journal of Pediatric Gastroenterology and Nutrition. We also draw on the 2022 hyperbilirubinemia guideline of the American Academy of Pediatrics, the AAP, written by Kemper and colleagues; the 2025 AAP clinical report on identifying infants with biliary atresia by two to four weeks of life, written by Harpavat and colleagues; the 2022 position paper of the Italian Society of Paediatric Gastroenterology, Hepatology and Nutrition, known as SIGENP; and the guideline on neonatal jaundice from the National Institute for Health and Care Excellence in England, known as NICE. The American Association for the Study of Liver Diseases and the European Association for the Study of the Liver, AASLD and EASL, have no guideline on neonatal cholestasis.

## Part one. Three clocks in the same infant.

Conjugated hyperbilirubinemia is a sign, not a diagnosis, and it is shared by several groups of disease that run on very different time scales. It helps to think of three clocks, with a fourth hazard that can strike at any time.

The first clock belongs to systemic and treatable disease: bacterial sepsis, urinary tract infection, galactosemia, tyrosinemia, and failure of the adrenal glands or the pituitary. Here what runs out is life itself, or the brain, through low blood sugar or bleeding, or the synthetic function of the liver. This clock is measured in hours to days.

The second clock belongs to obstruction, above all biliary atresia, but also a choledochal cyst or an obstructing stone. Here what runs out is the chance of draining bile through the porta hepatis while the liver is still soft enough to recover. This clock is measured in weeks, and, as we will see, every fortnight counts.

The third clock belongs to hepatocellular and genetic disease, such as alpha-1 antitrypsin deficiency, Alagille syndrome, the progressive familial intrahepatic cholestasis syndromes, and the bile acid synthesis defects. Here what runs out is mostly nutrition, growth and fat-soluble vitamins, and the clock is measured in weeks to months.

The fourth hazard is bleeding from vitamin K deficiency, which can complicate any form of cholestasis and can cause an intracranial hemorrhage at any visit.

Why does infection cause cholestasis at all? The mechanism is worth understanding step by step. Bacterial endotoxin and inflammatory cytokines, especially tumor necrosis factor, act on the hepatocyte and reduce the number and activity of the transporters that pump bile salts and conjugated bilirubin across the canalicular membrane into bile. Bile flow falls, conjugated bilirubin backs up into the blood, and the infant becomes jaundiced, sometimes before there is any fever. Galactosemia, tyrosinemia and adrenal or pituitary failure injure the hepatocyte through toxic metabolites or through the loss of hormones that the immature liver needs for normal bile formation. In all of these, treating the cause usually clears the cholestasis.

The practical rule that follows is simple. The sick-looking infant is the emergency. The well-looking infant is the one who may have biliary atresia.

## Part two. One milligram, regardless of total.

The 2004 guideline from the North American society, written by Moyer and colleagues, defined cholestasis as a direct bilirubin above one milligram per deciliter when the total bilirubin was below five, or above twenty percent of the total when the total was higher than five. The 2017 joint guideline dropped the twenty percent rule. It now says that a direct or conjugated bilirubin above one point zero milligrams per deciliter, which is about seventeen micromoles per liter, is abnormal regardless of the total bilirubin. So a direct bilirubin of one point two with a total of twelve is cholestasis, not breast milk jaundice, even though the fraction is only ten percent.

When should we test? The guideline says that a formula-fed infant who is still jaundiced at two weeks of age should have a total and a direct or conjugated bilirubin measured. A breastfed infant

who looks well, has a normal examination, and has normally pigmented stools may be observed until three weeks, and tested then if still jaundiced. The 2022 AAP hyperbilirubinemia guideline uses the same triggers, two weeks for formula-fed infants and three to four weeks for breastfed infants. A pale stool at any age means testing now, without waiting for these dates.

The guideline also reminds us that jaundice becomes visible at a total bilirubin of about two and a half to three milligrams per deciliter, and that eye estimates of how yellow an infant is are unreliable. Cholestatic jaundice affects about one in every two thousand five hundred term infants. That is rare for a general paediatrician but common enough that every busy practice will meet it several times a year.

## Part three. Ten recommendations, and the order of steps.

The 2017 guideline makes ten recommendations, and most carry a strong grade with high-quality evidence, printed as one A. In plain words, they say the following. Fractionate the bilirubin in formula-fed infants jaundiced at two weeks, and in well breastfed infants by three weeks. Always fractionate, and treat a conjugated bilirubin above one milligram per deciliter as pathological. Examine carefully for an enlarged liver or spleen and for an ill appearance. Look at the stool yourself. Use ultrasound to exclude a choledochal cyst and stones, remembering that an absent or abnormal gallbladder suggests atresia but does not prove it. Understand that a hepatobiliary scintigraphy scan, often called a HIDA scan, has limited specificity, although clear excretion of tracer into the bowel helps to exclude atresia. Do not rely on magnetic resonance cholangiography, endoscopic retrograde cholangiography or percutaneous cholangiography. Treat a well-timed liver biopsy read by an expert pathologist as the most supportive single test. And regard the intraoperative cholangiogram, with histology of the duct remnant, as the gold standard.

Notice that nine of the ten recommendations are about biliary atresia. The day-one exclusions are written into the text and into the list of first-tier tests, not into a numbered recommendation. That is one reason they are sometimes forgotten.

From the text and the table of tests, the order of steps can be reconstructed as five stages. The first stage is the trigger: jaundice at two weeks in a formula-fed infant or three weeks in a breastfed infant, a pale stool at any age, or a previously raised direct bilirubin. The second stage is fractionation of the bilirubin the same day. The third stage is to look: take a history, examine the infant, look at the stool, find the newborn screening results, and ask the single most important question, does this infant look ill? The fourth stage is the first tier of tests, aimed at treatable disease, together with vitamin K and a fasting ultrasound. The fifth stage is review by a paediatric hepatologist within the same week, with second-tier tests, a liver biopsy, a cholangiogram if needed, and genetic testing.

The 2025 AAP clinical report adds an important rule for primary care. A direct or conjugated bilirubin of one milligram per deciliter or more is an urgent call to a paediatric gastroenterologist, ideally so that the infant is seen before thirty days of age. The report warns that further laboratory tests or an ultrasound should not delay that call, because a normal result can falsely reassure. In other words, the first tier is started by whoever finds the abnormal number, but it is never a reason

to postpone the referral.

## Part four. What to send on the first day.

The first-tier tests in the guideline fall naturally into four groups, each answering one question.

The first group asks how the liver and the blood are doing. It includes the total and direct bilirubin, the transaminases, alkaline phosphatase, gamma-glutamyl transferase, which is usually called GGT, albumin, the international normalized ratio, or INR, a full blood count, and the blood glucose.

The second group asks whether there is an infection. It includes a urinalysis and urine culture in every infant, and a blood culture if the infant is unwell, with a low threshold for a full sepsis evaluation.

The third group asks about metabolic and endocrine disease. It includes urine reducing substances, a review of the newborn screening results, and thyroid-stimulating hormone and thyroxine if the screening result is not readily available.

The fourth group asks about structure and genes. It includes a fasting abdominal ultrasound and an alpha-1 antitrypsin level and phenotype.

The second tier, done with the hepatologist, adds serum bile acids, cortisol, ammonia, lactate, red cell galactose-1-phosphate uridyltransferase activity, which is the enzyme missing in classical galactosemia and is called GALT, urine succinylacetone and organic acids, viral polymerase chain reaction tests, a sweat chloride test, a spine film, an echocardiogram, an eye examination, and a gene panel.

Of all these, four results change management on the first day: the glucose, the INR, the urine culture, and the urine reducing substances.

What is the work-up actually looking for? A systematic review by Gottesman and colleagues, published in 2015, pooled the final diagnoses of one thousand six hundred and ninety-two infants with conjugated hyperbilirubinemia. Idiopathic neonatal hepatitis accounted for twenty-six percent, and biliary atresia for almost exactly the same share, twenty-five point nine percent. Infection accounted for eleven and a half percent, parenteral nutrition for six point four percent, metabolic disease for four point four percent, alpha-1 antitrypsin deficiency for four point one percent, perinatal hypoxia for three point seven percent, and hypopituitarism or hypothyroidism together for two percent. If we add up infection, metabolic and endocrine causes, about eighteen percent, roughly one infant in six, has a cause that is treatable within days. Within the infections, cytomegalovirus made up about one third, bacterial sepsis about one quarter, and urinary infection with Escherichia coli about one tenth. The idiopathic group keeps shrinking as genetic testing expands; the guideline estimates that about a quarter of cholestatic infants have a monogenic disease.

## Part five. Sepsis and urinary infection.

Urinary infection in a young infant can be silent. In 2002, Garcia and Nager, writing in Pediatrics, described jaundice as an early, and sometimes the only, sign of urinary tract infection in infancy, often without fever. More recent data are striking. In a Taiwanese study published in 2022 by Chen and colleagues, six hundred and fifteen afebrile, otherwise well jaundiced infants under one month of age had a urine culture, and fourteen point three percent had significant bacteriuria. Escherichia coli made up forty-five and a half percent of the isolates, and only about twenty-two percent of those E. coli were sensitive to ampicillin.

There is an important caveat. Most of these infants had unconjugated or mixed jaundice, so the numbers probably overstate the rate of infection in true cholestasis. But the decision rests on an asymmetry. A urine culture is cheap and harmless. A missed E. coli infection can progress to sepsis, and, as the next section shows, it may be galactosemia announcing itself. So we send urine for culture and reducing substances on the first day, febrile or not.

## Part six. Galactosemia: when E. coli sepsis is a metabolic clue.

Classical galactosemia is caused by deficiency of the GALT enzyme. Galactose comes from lactose, the sugar in both breast milk and standard formula. When the enzyme is missing, galactose-1-phosphate accumulates in the liver, kidney, lens and brain. The infant becomes ill once milk feeds are established, with vomiting, poor feeding, jaundice, an enlarged liver and a coagulopathy, and sometimes cataracts. In 1977, Levy and colleagues, writing in the New England Journal of Medicine, showed that neonates with galactosemia are especially prone to E. coli sepsis, which was often fatal. The incidence of classical galactosemia is about one in forty to sixty thousand births.

Why can the newborn screen fail us here? There are three reasons. First, the result can take up to a week, and the infant may be ill before it arrives. Second, a red cell transfusion gives a falsely normal GALT activity, because the measured enzyme comes from the donor cells. Third, not every screening programme includes galactosemia. At the bedside, urine reducing substances are a useful clue, but only while the infant is taking lactose; they can be negative if the infant is not feeding or is on intravenous fluids. The confirmatory test is red cell GALT activity.

The action is the most important part. On suspicion alone, stop lactose and change to a soy-based or elemental formula, then confirm the diagnosis later. The international guideline for classical galactosemia, published in 2017 by Welling and colleagues for the Galactosemia Network, treats this dietary change as an emergency measure. A useful pearl: breast milk is full of lactose, so the breastfed infant with galactosemia is the most exposed, not the least.

## Part seven. When the pituitary is the liver disease.

Congenital hypothyroidism and combined pituitary hormone deficiency, including septo-optic dysplasia, can both present with cholestasis. The clues are recurrent low blood sugar, a small penis or undescended testes in a boy, roving eye movements or nystagmus suggesting optic nerve hypoplasia, and midline defects. A normal-looking genital examination does not exclude the diagnosis.

The trap is the newborn screen. Most programmes screen with thyroid-stimulating hormone alone, which detects primary hypothyroidism but misses central hypothyroidism, in which the pituitary does not make enough thyroid-stimulating hormone in the first place. So the guideline asks for a thyroid-stimulating hormone together with a free thyroxine, an early-morning cortisol, and, if hypopituitarism is suspected, magnetic resonance imaging of the brain and an eye review.

There is also a danger specific to the work-up. An infant without cortisol and growth hormone cannot defend the blood glucose during a fast, and may go into adrenal crisis under stress. The guideline therefore recommends that the abdominal ultrasound in such an infant be done without fasting, to avoid severe hypoglycaemia.

How long does this take to recognise? In a small series of eight infants published by Karnsakul and colleagues in 2007, cholestasis began at a mean of one point seven weeks of age, but hypopituitarism was not diagnosed until a mean of seven point six weeks, and every infant was cortisol-deficient. Five of the eight had septo-optic dysplasia. On cortisol and thyroxine replacement, the cholestasis resolved after a mean of nine point seven weeks. In an earlier Australian series by Ellaway and colleagues, seven of twenty children with congenital hypopituitarism presented with cholestatic jaundice. The lesson is that a normal newborn thyroid screen does not clear the pituitary. A glucose, a free thyroxine and a cortisol do.

## Part eight. Vitamin K before the diagnosis.

Vitamin K is fat-soluble, and its absorption depends on bile. Once bile flow falls, oral vitamin K becomes unreliable. A German study of one hundred and eight cases of late vitamin K deficiency bleeding between 1980 and 1990, published by Sutor and colleagues in 1995, found that at least thirty-seven percent had an underlying cholestasis. Fifty-eight percent presented with an intracranial hemorrhage, and nineteen percent died. Parenteral prophylaxis at birth reduced the risk about fourteen-fold, compared with about three-fold for an oral dose.

The guideline turns vitamin K into a diagnostic test. Check the INR, give parenteral vitamin K, and check the INR again. If it corrects, the problem was vitamin K deficiency from malabsorption. If it does not correct, or if it is out of proportion to the liver tests, think of gestational alloimmune liver disease, a metabolic disease such as tyrosinemia or galactosemia, or sepsis. In other words, the second INR is a test in its own right.

## Part nine. Imaging and tissue: suggestive, then decisive.

For the well-looking infant, the main task is to confirm or exclude biliary atresia. A fasting ultrasound is the first imaging test. It reliably excludes a choledochal cyst and stones. For atresia itself, a 2016 meta-analysis of twenty-three studies in the American Journal of Roentgenology found that the triangular cord sign, a band of fibrous tissue at the porta hepatis, had a pooled sensitivity of seventy-four percent and a specificity of ninety-seven percent. An abnormal gallbladder had a sensitivity of eighty-five percent and a specificity of ninety-two percent, and the combination of ultrasound features reached a sensitivity of ninety-five percent and a specificity of eighty-nine percent. A normal ultrasound therefore does not exclude atresia.

The HIDA scan works the other way. A meta-analysis cited in the guideline found that failure of tracer excretion had a sensitivity of almost ninety-nine percent for atresia, but a specificity of only about seventy percent. Non-excretion is common in neonatal hepatitis, bile duct paucity, prematurity and parenteral nutrition. Clear excretion helps to exclude atresia; non-excretion proves very little. And priming with phenobarbital for five days before the scan can itself delay the Kasai operation.

A liver biopsy read by an experienced paediatric pathologist gives the correct diagnosis in about ninety to ninety-five percent of cases. The typical features of atresia are bile ductular proliferation, bile plugs and portal fibrosis. But giant cells, the feature we associate with neonatal hepatitis, are present in twenty to fifty percent of atresia biopsies. Alpha-1 antitrypsin deficiency, cystic fibrosis and parenteral nutrition can all mimic atresia, and a biopsy taken too early can look falsely reassuring. Complications occurred in about one point seven percent overall, and about four point six percent of infants, in one series cited by the guideline. The cholangiogram is the final word, but even it can mislead in up to one fifth of cases, for example when a hypoplastic but patent biliary tree in Alagille syndrome or cystic fibrosis fails to fill.

Beyond the guideline, serum matrix metalloproteinase seven, or MMP-7, is an emerging blood test. In one Chinese cohort published in Hepatology in 2018 by Yang and colleagues, a cut-off of about fifty-three nanograms per milliliter gave a sensitivity of almost ninety-nine percent and a specificity of ninety-five percent. It is promising, but it is not part of the 2017 algorithm. The practical rule is this: when the biopsy and the clinical picture disagree, go to the cholangiogram, not to a second week of waiting.

## Part ten. Why the algorithm is in a hurry.

The Kasai portoenterostomy works best when it is done early. In the French national cohort published by Serinet and colleagues in 2009, six hundred and ninety-five children had a Kasai, at a median age of sixty days, which did not change between 1986 and 2002. Survival with the native liver was fifty-seven percent at two years, thirty-eight percent at five years, thirty-two percent at ten years, and twenty-eight and a half percent at fifteen years, and it fell step by step with each fifteen-day increase in the age at surgery. The authors estimated that operating on every child before forty-six days of age would avoid about six percent of all paediatric liver transplants in France each year.

The guideline adds that about seventy percent of infants clear their bile if the Kasai is done before sixty days, compared with fewer than a quarter after ninety days. In the United States, the mean age at Kasai was sixty-one days, and forty-four percent of operations took place after sixty days. These figures explain why the AAP now asks for referral before thirty days of age.

## Part eleven. What the guidelines say.

The North American and European documents agree on most things, but differ in a few that matter.

On the definition, the joint 2017 guideline and both AAP documents use a direct bilirubin above one milligram per deciliter at any total. The Italian SIGENP position paper of 2022 keeps the older

twenty percent rule, and the English NICE guidance asks for expert advice when the conjugated bilirubin is above twenty-five micromoles per liter. The verdict is that they differ.

On when to test, North America uses two weeks for formula-fed and three weeks for breastfed infants, and NICE defines prolonged jaundice as more than fourteen days in a term infant and more than twenty-one days in a preterm infant. These are aligned in practice.

On the day-one exclusions, both sides include urine culture, reducing substances, thyroid tests, glucose and the INR, and SIGENP adds cortisol and bile acids to its first tier. Again, aligned.

On the HIDA scan, the joint guideline keeps it as a test with limited specificity, while SIGENP says it is no longer part of the standard work-up. They differ.

On genetic testing, North America places a gene panel in the second tier with a low threshold, and SIGENP recommends early next-generation sequencing in parallel with the metabolic work-up once atresia is excluded. Aligned.

On the target age, the material difference appears. The 2025 AAP report names a target of referral before thirty days of age. The European documents ask for speed but give no number, and Serinet's figure of forty-six days is often quoted instead. Finally, AASLD and EASL have no guideline on neonatal cholestasis at all, which is a gap.

## Part twelve. Where the algorithm fails in practice.

Most delays come from a handful of predictable traps. The first is false reassurance: the idea that a breastfed infant must have breast milk jaundice, that yellow stools at three weeks exclude atresia, or that a normal ultrasound rules it out. The second is the blind spots of the newborn screen: thyroid-stimulating hormone screening misses central hypothyroidism, a transfusion gives a false-normal GALT, and the screen result may simply not be back. The third is test delay: priming for a HIDA scan, waiting for second-tier results before referring, or repeating a biopsy instead of moving to a cholangiogram. The fourth is missing the basics: no glucose check, no vitamin K, or fasting the infant with hypopituitarism for an ultrasound. The failure is almost never a missing test. It is a test sent without a deadline.

## Part thirteen. What to do differently in clinic.

Let us return to the two infants from the beginning. The well breastfed boy with a direct bilirubin of one point four and light yellow stools has cholestasis until proven otherwise. He needs first-tier tests the same day, parenteral vitamin K, and review by a hepatologist within the week, with the clock for biliary atresia already running. The sick girl who is vomiting, with a large liver, needs a glucose, an INR, a blood and urine culture, urine reducing substances and antibiotics now. If she grows E. coli, or her INR is high, stop lactose immediately and send a red cell GALT, without waiting for the screen. If an infant has low blood sugar, a small penis and a direct bilirubin of three, think of hypopituitarism, send a cortisol and free thyroxine, arrange brain imaging, scan the abdomen without fasting, and be ready to give hydrocortisone. If a five-week-old has acholic stools and no gallbladder on ultrasound, call the hepatologist and the surgeon today and skip the HIDA

scan. If the INR stays high after vitamin K, treat it as liver failure and look for gestational alloimmune liver disease, metabolic disease and viral infection, with early contact with a transplant centre. And if a preterm infant on parenteral nutrition has a rising direct bilirubin, remember that atresia is still possible; look at the stools, scan the liver, and keep a low threshold for biopsy.

The whole chapter can be held in two sentences. In the sick-looking infant, treat the systemic cause today. In the well-looking infant, assume biliary atresia until it is excluded, and do it before thirty days.

## Questions to test yourself.

Question one. What is the 2017 definition of cholestasis in an infant? Answer: a direct or conjugated bilirubin above one milligram per deciliter, about seventeen micromoles per liter, regardless of the total bilirubin. The twenty percent rule was dropped.

Question two. When should a breastfed infant who looks well be tested? Answer: if still jaundiced at three weeks of age, or at any age if the stools are pale.

Question three. Which four first-day results change management immediately? Answer: the glucose, the INR, the urine culture and the urine reducing substances.

Question four. Why might the newborn screen miss galactosemia? Answer: the result can take up to a week, a red cell transfusion gives a falsely normal GALT, and some programmes do not screen for it.

Question five. Why might the newborn screen miss hypopituitarism? Answer: most programmes measure thyroid-stimulating hormone alone, which misses central hypothyroidism. A free thyroxine and a cortisol are needed.

Question six. Why should the infant with suspected hypopituitarism have a non-fasting ultrasound? Answer: because without cortisol and growth hormone the infant cannot maintain the blood glucose during a fast, and severe hypoglycaemia or adrenal crisis can follow.

Question seven. What does an INR that fails to correct after parenteral vitamin K suggest? Answer: true synthetic liver failure, from gestational alloimmune liver disease, a metabolic disease, or sepsis, rather than simple vitamin K malabsorption.

Question eight. What did the French national cohort show about the timing of the Kasai operation? Answer: native-liver survival fell with each fifteen-day delay, and operating on every infant before forty-six days would avoid about six percent of paediatric liver transplants each year.

End of chapter.